

Executive Summary

Retrieval-Augmented Generation (RAG) integrates a retrieval module that fetches external documents with a generative language model that produces text conditioned on the retrieved evidence. This architecture enables efficient fine-tuning and dynamic updating of the model's knowledge base without retraining the entire generator.

Key Findings

- Definition ? RAG couples information retrieval and text generation, allowing the system's internal knowledge to be altered efficiently through the retrieval component rather than full model re-training.
- Primary Challenge ? The most frequently cited vulnerability is RAG poisoning, where malicious or low-quality documents inserted into the retrieval corpus can corrupt the generated output.

Methodology Analysis

The core methodology of RAG consists of two stages:

Stage	Function	Typical Implementation
Retrieval	Searches a large external corpus (e.g., vector store, knowledge base) for passages relevant to the input query.	Dense vector similarity search, BM25, or hybrid approaches.
Generation	Conditions a language model on the retrieved passages to produce a response.	Encoder-decoder or decoder-only transformers fine-tuned on concatenated query-passage inputs.

Key properties derived from the provided answers:

1. Modular Knowledge Update ? Because the knowledge resides in the retrieval index, updating the corpus instantly reflects in generated answers without re-training the generator.
2. Fine-tuning Flexibility ? The generator can be fine-tuned on downstream tasks while still leveraging the external knowledge source, preserving adaptability.

Limitations and Future Directions

Limitation	Description	Potential Research Direction
RAG Poisoning	Insertion of adversarial or erroneous documents into the retrieval store can lead to misleading or harmful generations.	Develop robust corpus sanitization, provenance tracking, and retrieval-time verification mechanisms.
Retrieval Quality Dependency	Generation quality is bounded by the relevance and accuracy of retrieved passages.	Explore hybrid retrieval (semantic + lexical), dynamic relevance feedback, and multi-hop retrieval strategies.
Scalability of Indexes	Large corpora demand efficient indexing and low-latency search.	Investigate hierarchical or compressed vector indexes and hardware-accelerated search.
Alignment Between Retriever and Generator	Mismatch in representation spaces can reduce the usefulness of retrieved text.	Joint training of retriever and generator embeddings to improve end-to-end coherence.

References

1. Prompting Guide ? Retrieval?Augmented Generation.
<https://www.promptingguide.ai/techniques/rag>

2. NVIDIA Blog ? What Is Retrieval?Augmented Generation. <https://blogs.nvidia.com/blog/what-is-retrieval-augmented-generation>

3. Wikipedia ? Retrieval?augmented generation. https://en.wikipedia.org/wiki/Retrieval-augmented_generation

4. Pinecone Learn ? Retrieval?Augmented Generation. <https://www.pinecone.io/learn/retrieval-augmented-generation>

RAG Pipeline Evaluation Results

Question	MRR@10	nDCG@10	Precision@5	Recall@10	Faithfulness	Context Precision	Answer Relevancy
What is retrieval augmented generation?	-	-	-	-	-	-	-
What are common challenges with RAG systems?	-	-	-	-	-	-	-